Cash Flow, $							
Year:	**0**	**1**	**2...**	**... t-1**	**t**	**t + 1...**	**Present Value**
Perpetuity		1	1...	1	1	1...	$\frac{1}{r}$
t-period annuity		1	1...	1	1		$\frac{1}{r}-\frac{1}{r(1+r)^t}$
t-period annuity due	1	1	1...	1			$(1+r)\left(\frac{1}{r}-\frac{1}{r(1+r)^t}\right)$
Growing perpetuity		1	$1\times(1+g)$ ...	$1\times(1+g)^{t-2}$	$1\times(1+g)^{t-1}$	$1\times(1+g)^{t}$ ...	$\frac{1}{r-g}$
t-period growing annuity		1	$1\times(1+g)$ ...	$1\times(1+g)^{t-2}$	$1\times(1+g)^{t-1}$		$\frac{1}{r-g}\times\left(1-\frac{(1+g)^t}{(1+r)^t}\right)$

Useful maths

Mean and StDev

For outcomes x_i with probabilities p_i

$$\mu_x = \sum_i x_i\, p_i$$

$$\sigma_x = \sqrt{\frac{1}{N}\sum_i^N (x_i-\mu_x)^2}$$

Continuous random variables with pdf $p(x)$

$$\mu_x = \int x\, p(x)$$

$$\sigma_x = \sqrt{\int (x-\mu_x)^2\, p(x)dx}$$

Sample standard deviation

With a sample x_i, approximation $\sigma \approx s$

$$s = \sqrt{\frac{1}{N-1}\sum_{i=1}^N (x_i-\mu_x)^2}$$

Covariance

$$Cov(x,y) = \sum_i (x_i-\mu_x)(y_i-\mu_y)\, p_i = \sigma_{xy} = \sigma_x\sigma_y\rho_{xy}$$

Correlation

Pearson product-moment correlation coefficient

$$\rho_{xy} = r_{xy} = \frac{\sigma_{xy}}{\sigma_x\sigma_y} = \frac{\sum^N (x-\bar{x})(y-\bar{y})}{\sqrt{\sum^N (x-\bar{x})^2(y-\bar{y})^2}}$$

Least Squares

Variables x, y have least squares slope

$$m = r_{xy}\frac{\sigma_x}{\sigma_y}$$

Intercept

$$c = \mu_y - m\mu_x$$

General propagation of uncertainty

For $f(x)$, with σ_x at a particular value x_0

$$\sigma_f(x_0) \approx \sigma_x \left|\frac{df}{dx}\right|_{x_0}$$

For $f(x,y)$, with covariance $\sigma_{xy} = \sigma_x\sigma_y\rho_{xy}$

$$\sigma_f^2(x_0,y_0) \approx \sigma_x^2\left|\frac{\partial f}{\partial x}\right|^2 + \sigma_y^2\left|\frac{\partial f}{\partial y}\right|^2 + 2\sigma_{xy}\frac{\partial f}{\partial x}\frac{\partial f}{\partial y}$$

Quadratic formula (*Sanity check*)

$$ax^2+bx+c=0$$

$$\Rightarrow x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$$

Taylor

$$f(x) \equiv \sum_{n=0}^{\infty}\frac{f^{(n)}(a)}{n!}(x-a)^n$$

L'Hôpital

[Often] when $f(x\to c) = g(x\to c) = 0$ or $\pm\infty$,

$$\lim_{x\to c}\frac{f(x)}{g(x)} = \lim_{x\to c}\frac{f'(x)}{g'(x)}$$

Confidence intervals

For a confidence interval CI%

$$\text{Q-value} = NORMSINV\left[1-\frac{1-CI\%}{2}\right]$$

So that the confidence interval

$$CI = Mean \pm Q\times\sigma$$

Gaussian PDF

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma}e^{-\frac{(x-\mu)^2}{2\sigma^2}}$$

Geometric average

Returns (e.g. time series) $\mu_{i\in N}$, compounding average

$$\begin{aligned} Geometric\ mean &= \left(\prod_{i=1}^N (1+\mu_i)\right)^{1/N} \\ &= \sqrt[N]{(1+\mu_1)(1+\mu_2)\ldots(1+\mu_N)} \end{aligned}$$

Options and risk management

Sell call: "You buy this stock in future at a set price"
Buy call: "I can buy this stock at set price in future"
Sell put: "You can sell this stock to me at a set price in future"
Buy put: "I can sell this stock to you at set price in future"

IR swaps

Lay off risk: Pay fixed and receive floating in IR swap (-ve duration)

Party that receives fixed has duration equivalent to owning the bond (+ve). Party that pays has -ve. Change in equity ΔP for a change in yield is backed out from duration of bond with price P

$$Dur = \frac{\left(\frac{\Delta P}{P}\right)}{\left(\frac{\Delta(1+y)}{1+y}\right)}$$

Binomial pricing:

Portfolio is perfectly hedged if one holds H shares of stock for each call option written, where C and S are the prices in future outcomes:

$$H = \frac{C_u - C_d}{S_u - S_d}$$

Equivalent portfolio is holding (1-H) shares and placing remaining funds (from stock + put/call portfolio) into T-bills

Black Scholes

$$\begin{aligned} \textit{Value of call option} &= [\text{delta} \times \text{share price}] - [\text{bank loan}] \\ &= [N(d_1) \times P] - [N(d_2) \times EXe^{-rT}] \end{aligned}$$

where

$$d_1 = \frac{\ln[P/EX] + (r + \sigma^2/2)T}{\sigma\sqrt{T}};$$
$$d_2 = d_1 - \sigma\sqrt{T};$$

$N(d) = NORMSIDST(d)$;
$EX =$ Exercise price of option;
$T =$ number of periods to expiration;
$P =$ stock price now; and
$\sigma =$ standard deviation per period of (continuously compounded) rate of return of stock
r = continuously compounded risk free rate

B-S With dividends

r_d = annual dividend yield

$$\textit{Value of call option} = [N(d_1) \times S_0 e^{-r_d T}] - [N(d_2) \times EXe^{-rT}]$$

$$\textit{Value of put} = EXe^{-rT}[1 - N(d_2)] - S_0 e^{-r_d T}[1 - N(d_1)]$$

where

$$d_1 = \frac{\ln[P/EX] + (r - r_d + \sigma^2/2)T}{\sigma\sqrt{T}};$$
$$d_2 = d_1 - \sigma\sqrt{T};$$

Put call parity (ex-dividend S_0)

$$P_0 = C_0 + PV(X) - S_0$$

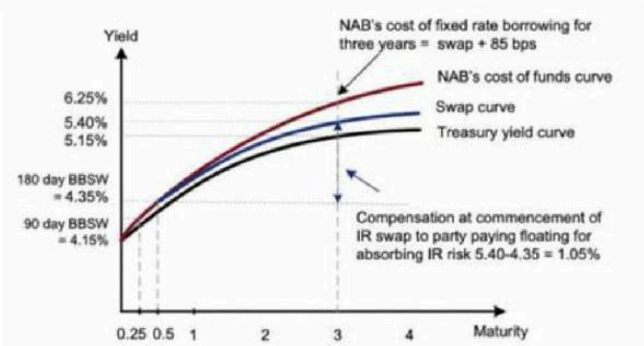
Factor increase	Call value	Put value
Stock price	Increases	Decreases
Exercise price	Decreases	Increases
Volatility	Increases	Increases
Time to expiration	Increases	Increases
Interest rate	Increases	Decreases
Dividend rate	Decreases	Increases

Hedge ratio or *delta*

The number of stocks required to hedge against the price risk of holding one option (d_1 from B-S)

$$Call = N(d_1)$$
$$Put = N(d_1) - 1$$
$$delta = \frac{Change\ in\ the\ value\ of\ the\ option}{Channge\ in\ the\ value\ of\ the\ stock}$$

Treasury yield curve, swap curve and funding curve

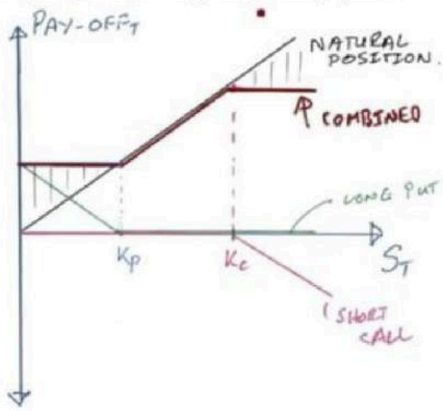


The cost to NAB of fixing the interest rate for three years is an extra 125 bps per year (5.40%-4.15%) at the time the bonds are issued. The extra 125 bps buys NAB protection against interest rate rises over the three years.

Collars

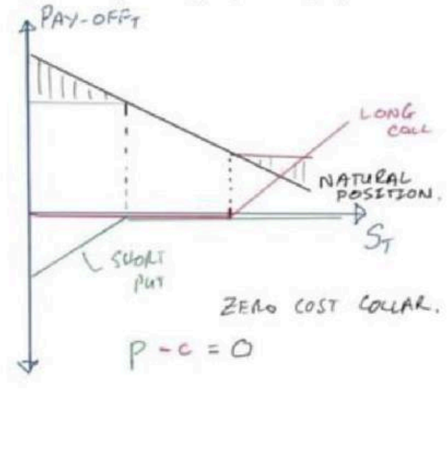
Natural long positions (e.g. Gold miner):

- Payoff increase with increase of asset price
- Protect downside with long put
- Pay for put by selling call, limiting upside



Natural short positions (e.g. Airline in oil):

- Payoff decrease with increase of asset price
- Protect downside with long call
- Pay of call by selling put, limiting upside



Markets

Synthetic bonds / law of one price

Consider three cash flows C_1, C_2 and C_3. If
$$C_3 = A \times C_1 + (1-A) \times C_2,$$
then
$$A = \frac{C_3 - C_2}{C_1 - C_2},$$
with
$$PV_3 = A \times PV_1 + (1-A) \times PV_2$$
e.g. C_1=\$100, $C_2 = \$0$ and $C_3 = \$50$, Then
$$A = \frac{\$50 - \$0}{\$100 - \$0} = 50\%$$

Two-asset portfolio

Mean returns $\mu_{A,B}$, risk (StDev) $\sigma_{A,B}$, correlation
$$\rho_{AB} = \frac{cov(\mu_A, \mu_B)}{\sigma_A \times \sigma_B}$$
risk-free rate r_f. Weights ω_A and $\omega_B = 1 - \omega_A$.

Mean portfolio return
$$\mu_p = \omega_A \times \mu_A + \omega_B \times \mu_B$$

Variance of portfolio
$$\begin{aligned}\sigma_p^2 &= \omega_A^2\,\sigma_A^2 + \omega_B^2\,\sigma_B^2 + 2\,\omega_A\,\omega_B\,cov(r_A, r_B)\\ &= \omega_A^2\,\sigma_A^2 + \omega_B^2\,\sigma_B^2 + 2\,\omega_A\,\omega_B\,\rho_{AB}\sigma_A\sigma_B\end{aligned}$$

Minimum-variance portfolio

$$\omega_A = \frac{\sigma_B^2 - cov(r_A, r_B)}{\sigma_A^2 + \sigma_B^2 - 2\,cov(r_A, r_B)}$$

Optimal (tangent) portfolio is given by A-stock weighting:
$$\omega_A = \frac{(\mu_A - r_f)\sigma_B^2 - (\mu_B - r_f)\rho_{AB}\sigma_A\sigma_B}{(\mu_A - r_f)\sigma_B^2 + (\mu_B - r_f)\sigma_A^2 - (\mu_A + \mu_B - 2r_f)\rho_{AB}\sigma_A\sigma_B}$$

Sharpe ratio / 'reward to risk' is
$$Sharpe\ ratio = \frac{\mu_p - r_f}{\sigma_p}$$

** Sharpe is max. along Tangent line/**Capital Market Line** ('CML')*
$$\mu_p = Sharpe \times \sigma_p + r_f$$

For CML portfolio with risky asset allocation $\omega_p = 1 - \omega_f$
$$\mu_C = (1 - \omega_p)r_f + \omega_p\mu_p$$
$$\sigma_C = \omega_p\sigma_p$$

To achieve a target return μ_T,
$$\omega_p = \frac{\mu_T - r_f}{\mu_p - r_f}$$

Utility

$$U = E(r) - 0.5A\sigma^2$$
A>0: Risk aversion, preference for lower risk
A=0: Risk neutral, highest return sought
A<0: Pathological risk seeking, lower returns ok with ↑ risk

With risk-free rate r_f, and risky asset r_p, σ_p, maximum level of risk aversion for which risky asset is preferred:
$$A = \frac{2(r_p - r_f)}{\sigma_p^2}$$

For the CML portfolio, utility is
$$U = (1 - \omega_p)\,r_f + \omega_p\mu_p - \frac{1}{2}A\omega_p^2\sigma_p^2$$
So to maximise utility,
$$\frac{dU}{d\omega_p} = -r_f + \mu_p - A\omega_p\sigma_p^2 \to 0$$
$$\Rightarrow \omega_p = \frac{\mu_p - r_f}{A\sigma_p^2}$$

Capital Asset Pricing Model / CAPM

$$Market\ risk\ premium = r_m - r_f$$

Required return on risky asset (systematic risk only!)
$$r = r_f + \beta \times (r_m - r_f)$$

Asset Beta:
$$\beta = \frac{cov(stock, market)}{var(market)}$$
$$\beta > 1 \Rightarrow \text{Aggressive stock}$$
$$\beta < 1 \Rightarrow \text{Defensive stock}$$

Realised CAPM in terms of risk premium: *R=r-rf*, α is anomalous return and e_i is random term
$$R_i = \alpha_i + \beta_i R_M + e_i$$

Risk is expressed in terms of systematic and firm specific components of variance
$$\sigma_i^2 = \beta_i^2\sigma_M^2 + \sigma^2(e_i)$$

With R-square, expressing portion of variation in stock movement that is systematic:
$$\rho^2 = \frac{\beta_i^2\sigma_M^2}{\sigma_i^2}$$

When there is a portfolio of stocks P with weight ω, so that
$$R_P = \alpha_P + \beta_P R_M + e_P$$
$$\beta_P = \sum_i \omega_i\beta_i$$
$$\alpha_P = \sum_i \omega_i\alpha_i$$
$$e_P = \sum_i \omega_i e_i$$
$$\sigma_P^2 = \beta_P^2\sigma_M^2 + \sigma^2(e_P)$$

And for two stocks *i* and *j*
$$\mathrm{Cov}(R_i, R_j) = \beta_i\beta_j\mathrm{Cov}(R_M, R_M) = \beta_i\beta_j\sigma_M^2$$
$$\rho_{ij} = \frac{\beta_i\beta_j\sigma_M^2}{\sigma_i\sigma_j}$$

Fama-French

CAPM, but with three betas for market, firm size and book-to-market ratios.
$$\begin{aligned}\alpha \;=\; r_f \;&+\; \beta_{market} \times (r_m - r_f)\\ &+\; \beta_{size} \times (r_{small} - r_{large})\\ &+\; \beta_{BtM} \times (r_{BtM\uparrow} - r_{BtM\downarrow})\end{aligned}$$

$$\alpha > 0 \Rightarrow +\text{ve risk-adjusted returns}$$
$$\alpha < 0 \Rightarrow -\text{ve risk-adjusted returns}$$

Debt

Effective annual rate

With a compounding period T (measured in years e.g. 1 month $T = 1/12$), return over that period $r_f(T)$

$$\begin{aligned} EAR &= \left[1 + r_f(T)\right]^{1/T} - 1 \\ &= (1 + APR \times T)^{1/T} - 1 \\ &= \exp(r_{cc}) - 1 \end{aligned}$$

Annual percentage rate: $APR = r_f(T) \times (1/T)$

Continuously compounding rate: $r_{cc} = \ln(1 + EAR)$

Real and nominal interest rates

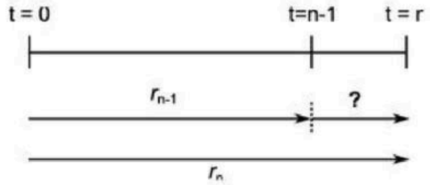
i=inflation

$$r_{real} = \frac{r_{nom} - i}{1 + i}$$

$$i = \frac{r_{nom} - r_{real}}{r_{real} + 1}$$

$$r_{real} \approx r_{nom} - i$$

Forward rates



$$\text{n-year forward rate} = \frac{(1 + r_n)^n}{(1 + r_{n-1})^{n-1}} - 1$$

Forward rates give expectation of interest rates, as short-rates based on yields to maturity of different-duration assets. Discount future cashflows by **product** of relevant forward rates (taken from t=0, these products are just the compounded yields).

Liquidity preference theory

This theory asserts that forward rates reflect expectations about future interest rates plus a liquidity premium that increases with maturity:

Interest rate = Forward Rate – Liquidity Premium

Deferred loans

Implied annually compounded forward rate for a deferred loan of length X beginning in year Y, based on the yields r is

$$f = \left[\frac{(1 + r_{X+Y-1})^{X+Y-1}}{(1 + r_{Y-1})}\right]^{\frac{1}{X}}$$

Bond equivalent ytm

Convention: work out semi-annual yield to maturity, (i.e. 2x number of periods) then double it.

Realised compounding yield to maturity

= yield with reinvestment of coupons at a given interest rate. Add up total proceeds, including T-compounded value of all coupons.

$$Real.ytm = \left(\frac{Total\ proceeds}{Current\ price}\right)^{1/T} - 1$$

Equivalent annual costs (Loan repayments)

Take out a loan of NPV=$***L***, paid over a period of ***n*** with interest rate ***r***. ***n***-year annuity factor is

$$AF_n = \frac{1}{r} - \frac{1}{r(1+r)^n}$$

Equivalent per-period costs (repayments) is

$$C = \frac{NPV}{AF_n}$$

NPV

$$NPV = -CF_0 + \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \cdots + \frac{CF_T}{(1+r)^T}$$

IRR: Expected rate of return. Use IRR({Cashflows}) in excel

$$IRR \equiv r \Leftrightarrow NPV = 0$$
$$IRR > discount \Rightarrow accept$$
$$IRR < discount \Rightarrow reject$$

Bonds: YTM ≡ IRR. (BBB+ investor, BBB- junk)

$$\text{YTM} = \text{coupon} \Rightarrow \text{'Par'}$$

$$\text{YTM} < \text{coupon} \Rightarrow \text{'Premium'}$$

$$\text{YTM} > \text{coupon} \Rightarrow \text{'Discount'}$$

$$\text{YTM} \approx \text{RFR} + \text{Default risk} + \text{Interest rate risk} + \text{Premium for embedded options}$$

Interest rates ↑ ⇒ bond price ↓, Interest rates ↓ ⇒ bond price ↑

E.g. with coupons C, YTM r, num periods T, and face value F

$$\text{PV(coupon + FV)} = \text{PV}(r, T, C, FV)$$

$$\text{PV(coupon only)} = \text{PV}(r, T, C)$$

$$\text{PV(FV only)} = \text{PV}(r, T, 0, FV)$$

YTM = RATE(#periods, payments, PV, FV) or IRR({Cashflows})
*NOT THE DISCOUNTED CASHFLOWS, and with -ve PV

[**Check semi-annual**, for rates + yields. PV must be negative.]

$$Bid = sell\ \$\ to\ broker, Ask = buy\ \$\ from\ broker$$

$$Liquidity \propto 1/(Ask - Bid)$$

Duration

= weighted average of the times when the bond's cash payments are received.

$$\text{Duration} = \frac{1 \times \text{PV}(\text{C}_1)}{\text{PV}} + \frac{2 \times \text{PV}(\text{C}_2)}{\text{PV}} + \cdots + \frac{T \times PV(\text{C}_T)}{\text{PV}}$$

Portfolio duration

Total equity in portfolio = Σ Assets – Σ Liabilities

Portfolio duration = Multiply component durations by their values (liabilities negative), divide by total equity.

$$Portfolio\ duration = \frac{\sum Asset \times Durat. - \sum Liab. \times Durat.}{Total\ equity\ in\ portfolio}$$

Modified duration

= percentage change in bond price for a 1 percentage-point change in yield (adjusts for accuracy). Approximates % change in bond price for 1% change in yield. Units is 'years', so %Δ(price) = [yr] * [%Δrate/year] = %.

$$\text{Mod dur} = \text{volatility (\%)} = \frac{\text{Duration}}{1 + \text{yield}}$$

$$\text{Mod dur} = \frac{\left(\frac{\Delta P}{P}\right)}{\left(\frac{\Delta(1+y)}{1+y}\right)}$$

Trade discounts

Discount r_d for payment at t_1 days rather than usual t_2 day payment terms

$$\Rightarrow WACC = \left(\frac{1}{1 - r_d}\right)^{\frac{365}{t_2 - t_1}} - 1$$

Equity

Holding Period Return

For a stock of price S, with dividend D

$$E(r_1) = \frac{D_1 + S_1 - S_0}{S_0}$$

Dividend Yield

$$r_d = D_1/S_0$$

Gordon model

Assume constant growth of dividends g, req. rate of return r
(Last year's dividend D_0)
Next year's dividend $D_1 = D_0(1+g)$
Present value of future cash payments in perpetuity

$$P = \frac{D_1}{r-g}$$

$$r = r_f + \beta \times (r_M - r_f)$$

$$g = \text{Plowback ratio} \times \text{Return on equity}$$

$$\text{Payout ratio} = \frac{\text{Dividend per share}}{\text{Earnings per share}}$$

$$\text{Plowback ratio} = 1 - \text{Payout ratio}$$

$$\text{Return on equity} = \frac{\text{Earnings per share}}{\text{Book value per share}} = \frac{\text{Net income}}{\text{Book value of equity}}$$

$$\text{Earnings per share} = \frac{\text{Net income}}{\text{Number of outstanding shares}}$$

$$\text{Book value} = \text{Common stock} + \text{Retained earnings}$$

$$\text{Common stock} = \text{Face value of shares}$$

No-Growth Value

Dividend @ 100% payout

$$NGV = \frac{EPS_1}{r}$$

Present Value of Growth Opportunities

PVGO = P – NGV

$$PVGO = \frac{D_0(1+g)}{r-g} - \frac{E_1}{r}$$

PE Ratios with growth

$$P_0 = \frac{D_1}{r-g} = \frac{E_1(1-b)}{r - (b \times ROE)}$$

$$\frac{P_0}{E_1} = \frac{1-b}{r - (b \times ROE)}$$

(b = plowback ratio)

Residual dividend approach

Project returns: r_p
Shareholder cost of equity: r_e
If $r_p > r_e$, company creates value for shareholders
If $r_p = r_e$, shareholders indifferent
If $r_p < r_e$, company destroys value for shareholders

**Stipulate whether average or year-end equity is used*

Gordon two-stage

(Last year's dividend D_0)
e.g. 3 years' variable growth g_1, g_2 and g_3, after which growth becomes constant g_4.

Dividends are

$$D_1 = D_0(1+g_1)$$
$$D_2 = D_1(1+g_2)$$
$$D_3 = D_2(1+g_3)$$
$$D_4 = D_3(1+g_4) \text{ [Caution!]}$$

Present value of future constant growth at t = 3 is

$$P_3 = \frac{D_4}{r - g_4}$$

Stage 1: Sum of discounted D_1, D_2 and D_3 to t=0

$$S_1 = \sum_{t=1}^{3} \frac{D_t}{(1+r)^t}$$

Stage 2: Discount P_3 to t=0

$$S_2 = \frac{P_3}{(1+r)^3} = \frac{D_3(1+g_4)}{(r-g_4)(1+r)^3}$$

$$PV = Stage\ 1 + Stage\ 2$$

EBITDA multiples

$$\text{EBITDA Multiple} = \frac{\text{EV}}{\text{EBITDA}} \approx \frac{1}{r-g}$$

↓ multiple ⇒ ↑ r (risk) or ↓ g (growth)

Calculate multiples for comparable companies, to work out enterprise value

$$\text{Enterprise value} = \text{EBITDA} \times \text{Multiple}$$

$$\text{Fair val. of Equity} = \text{EV} + \text{excess cash} - \text{long term debt}$$

$$\text{Fair value} = \frac{\text{Equity}}{\text{\# shares outstanding}}$$

Residual income model

PV of stock price in terms of ROE and book value of firm equity (per share) projected. PV increases when ROE>r_e:

$$PV = BV_0 + \frac{(ROE_1 - r_e)BV_0}{(1+r_e)} + \frac{(ROE_2 - r_e)BV_1}{(1+r_e)^2} + \frac{(ROE_3 - r_e)BV_2}{(1+r_e)^3} + \cdots$$

Dupont

<table>
<tr><td colspan="4">$ROE = \frac{Net\ income}{Shareholder\ equity}$</td></tr>
<tr><td colspan="4">How effectively has the firm been run to benefit shareholders</td></tr>
<tr><td colspan="2">Return from ops for shareholders + creditors
$ROA = \frac{NOPAT = NI + int.exp. \times (1-T)}{Average\ assets}$</td><td colspan="2">How well did the firm use debt financing to benefit shareholders</td></tr>
<tr><td>Asset turnover
$\frac{Sales}{Av.assets} \times$
"How much revenue from assets?"</td><td>Operating efficiency
$\frac{NOPAT}{sales} \times$
"What portion of each $ revenue is left for creditors + shareholders after covering operating expenses?"</td><td>Interest efficiency
$\frac{N.I.}{NOPAT} \times$
"What portion of each NOPAT $ is left after covering interest expense?"</td><td>Leverage
$\frac{Av.assets}{Av.sh.equity}$
"What assets beyond equity did the company hold?"</td></tr>
</table>

Capital structure

Cost of debt

$$r_d = r_f + credit\ spread$$

Credit spread = default risk for firms of similar *interest coverage*

$$IC = \frac{EBIT}{Interest\ expenses}$$

Cost of equity (From CAPM):

$$r_e = r_f + \beta \times Market\ risk\ premium$$

Weighted Average Cost of Capital (WACC)

Financial + systematic risk: $V = D + E$

$$WACC = (1 - T_c) \times \frac{D}{V} \times r_d + \frac{E}{V} \times r_e$$

Levered/unlevered return (concentration of risk)

r_E^U is return on unlevered equity = **return on assets**, leverage D/E, gives r_E^L as cost of levered equity, and:

$$r_E^L = r_E^U + \left(\frac{D}{E}\right)(1 - T_c)(r_E^U - r_D)$$

Levered/unlevered beta

When determining cost of equity, consider *current* firm leverage D_c/E_c vs *target* leverage D_t/E_t. Levered beta = business risk + financial risk.

$$\beta_L = \beta_U \times \left[1 + \frac{D}{E} \times (1 - T_c)\right]$$

$$\beta_U = \frac{\beta_L}{\left[1 + \frac{D}{E} \times (1 - T_c)\right]}$$

So, to determine *target* beta:

$$\beta_t = \beta_c \times \frac{1 + \frac{D_t}{E_t}(1 - T_c)}{1 + \frac{D_c}{E_c}(1 - T_c)}$$

Tax shields

When value of debt *D* is permanent, tax shields with deductions at tax rate *T* increases firm value:

$$\text{Value of levered firm} = \text{Value if all equity-financed} + T_c \times D$$

Leverage => add financial distress (measure as PV, -ve). Optimise!

Value of government claim on firm

Tax rate T_c, PV of taxes paid is

$$PV(T) = (E + T) \times T_c = \frac{E \times T_c}{1 - T_c}$$

Sustainable Growth Rate (SGR)

Rate at which firm can grow in without further equity raise, assuming new debt can be taken on. (Uses **book value** of equity)

$$\begin{aligned} SGR &= \frac{\text{Retained earnings}}{\text{Beginning equity}} \\ &= \frac{R.E.}{N.I.} \times \frac{N.I.}{Sales} \times \frac{Sales}{N.A.} \times \frac{D + E}{E} \\ &= Plowback \times N.Margin \times Asset\ util. \times \left(1 + \frac{D}{E}\right) \end{aligned}$$

Net assets = BV.E. + Liabilities(Loans)

Buybacks

ΔDebt = excess cash used + new debt raised

$$PV\ (Tax\ shields) = \Delta D \times T_c$$

Debt shield *Value to share sellers* = Premium × Volume

% to sellers = Value to sellers / PV (Tax)

$$E_{new} = E_{old} - \Delta D + \Delta D \times T_c$$

Total Enterprise Value (TEV)

TEV = D + E – non-operating cashflows

During buybacks, incorporate changes including – cost of buyback + PV tax shields + cash value of new debt

Equate to PV of sustainable cash flows

$$TEV = \frac{EBIT \times (1 - T_c)}{WACC}$$

Metrics

Economic Value Added (Stern-Stewart)

$$\begin{aligned} EVA &= residual\ income = income\ earned - income\ required \\ &= income\ earned - cost\ of\ capital \times investment \end{aligned}$$

Economic Profit (McKinsey)

$$EP = (ROI - r) \times capital\ invested$$

Earnings per share

$$EPS = \frac{\text{Net income}}{\text{Number of shares outstanding}}$$

Return on equity (net less r_e)

$$ROE = \frac{\text{Net income}}{\text{Book value of common equity}}$$

Return on assets (net less WACC)

$$ROA = \frac{\text{Net operating profit after taxes (NOPAT)}}{\text{Total assets}}$$

Price-Earnings ratio

$$PE = \frac{\text{Share price}}{EPS}$$

Personal vs Corporate tax rates for dividends

Corp. tax rate T_c, personal tax rate T_p. Compare:

1. Distribute \$1 today, shareholder invest for 1 year will have
$$\$(1 - T_p) \times \left(1 + r \times \left(1 - T_p\right)\right)$$
2. Firm invest \$1 today, distribute next year, shareholder will have
$$\$(1 - T_p) \times \left(1 + r \times (1 - T_c)\right)$$

$T_c > T_p \Rightarrow payout \uparrow, T_c < T_p \Rightarrow payout \downarrow$

Franking credit

If T_c is corp tax rate and DIV is franked credit distributed, then franking credit is

$$FC = DIV \times \frac{T_c}{1 - T_c}$$

Dividend policy

$r_{proj} < r_e \Rightarrow$ poor projects, FCFE to shareholders: Dividends / buybacks, review policies/management

$r_{proj} > r_e \Rightarrow$ good projects, invest where cash available

Working capital management

ΔWorking capital = ΔAcc rec. + Δ Inv - Δ Acc payable

Work out PV cash flow perpetuity after tax for $r = WACC$

$$PV = (1 - T) \times \frac{\Delta EBIT}{r}$$

$$NPV = -\Delta WC - PV$$